

Our Experience of a left approach on Anterior Cervical Corpectomy and Fusion for Patient with Cervical Spondylitis Tuberculosis

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Introduction

CERVICAL SPINE TUBERCULOSIS (TB)

The cervical spine tuberculosis is again a rare condition with an occurrence rate of **2% to 5%** among spinal cases.

- The anterior approach to the cervical spine is commonly utilized for a variety of degenerative, traumatic, neoplastic, and infectious indications.
- We **report a case of tuberculosis of the cervical spine with neurological symptoms managed surgically.**
- The aim of surgical treatment is **to remove the pressure on the spinal cord and nerve roots**
- Patients operated on the anterior cervical region with an approach from the left to more easily reach the disc and to reduce the risk of complications

Case Report

A 46-year-old woman complained of **severe neck pain that radiated to the arms and legs in the last 2 months ago**. The pain felt increasingly aggravated with **weakness for both upper and lower extremity**. **Urination that could not be held in the last 1 month. Defecation cannot be controlled.**

The patient **has a history of long cough for 7 months and does not take anti-tuberculosis drugs**. There was no history of previous trauma.

Case Report

Physical examination revealed a **gibbus, tenderness, and limited range of motion at cervicothoracal region.**

Neurological examination revealed **a decreased motor strength in the upper and lower extremity and hypoesthesia below level C5.**

Xray & MRI examination revealed **a disruption of alignment of anterior and posterior vertebral body of cervical 5–6.**

Case Report

Pre Op

Neurological examination:

Pre op:

Motoric:

Shoulder elevation 4/4

Shoulder abduction 4/4

Elbow flexion 4/4

Wrist extension 4/4

Elbow extension 4/4

Wrist flexion 3/3

Finger flexion 3/3

Finger abduction 3/3

Hip flexion 1/3

Knee flexion 1/3

Knee extension 1/3

Ankle dorsiflexion 1/3

Ankle plantarflexion 1/3

Great toe extension 1/3

Great toe flexion 1/3

Sensoric:

Hypoesthesia below level C5

Physiological reflex:

Biceps: +/+

Triceps: +/+

Patella : +/+

ankle : +/+

Brachioradialis +/+

Pathological reflex:

Babinski -/-

Chaddock -/-

Oppenheim -/-

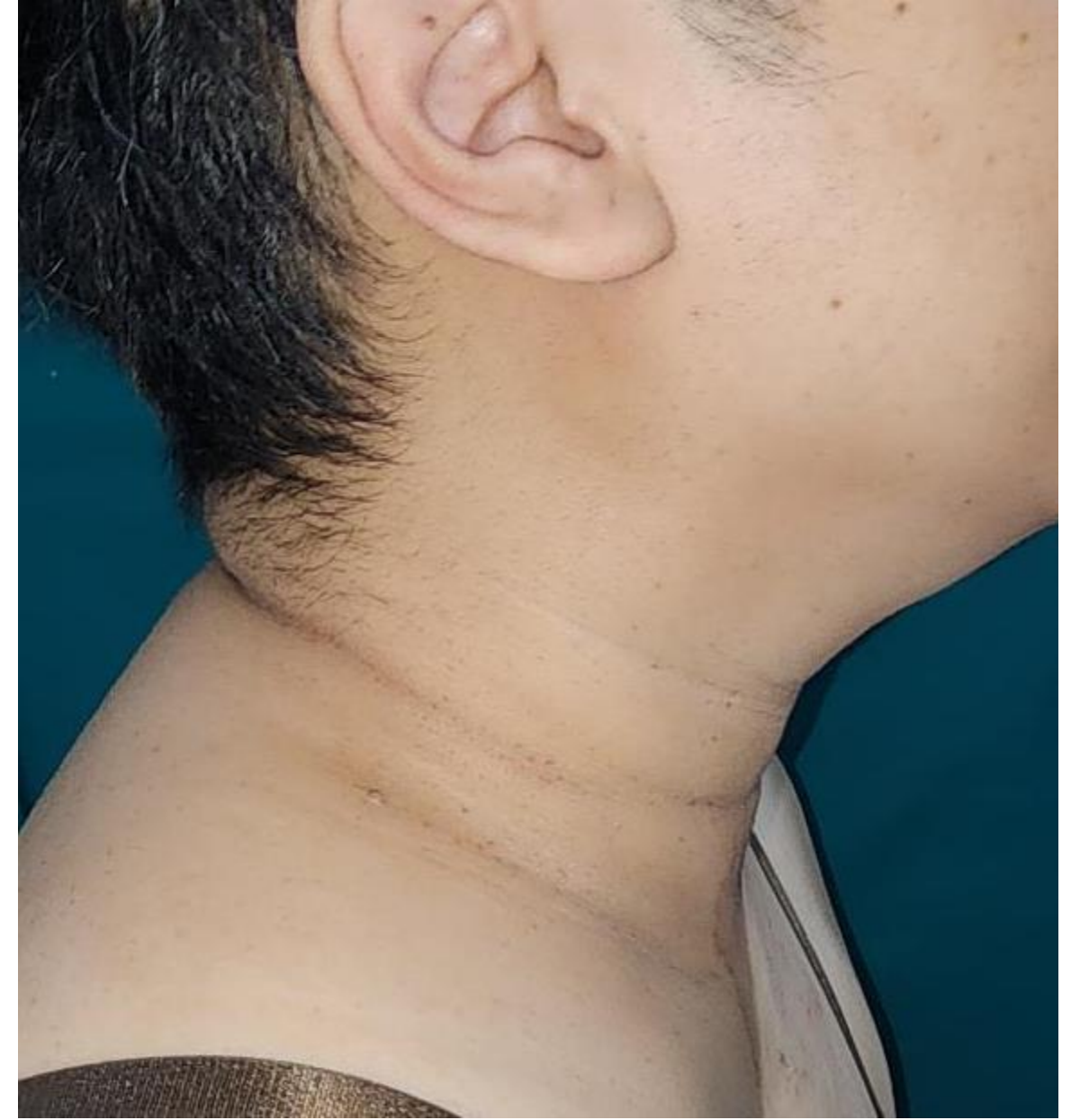
Clonus -/-

Vegetative:

BAK: on catheter

BAB: incontinence

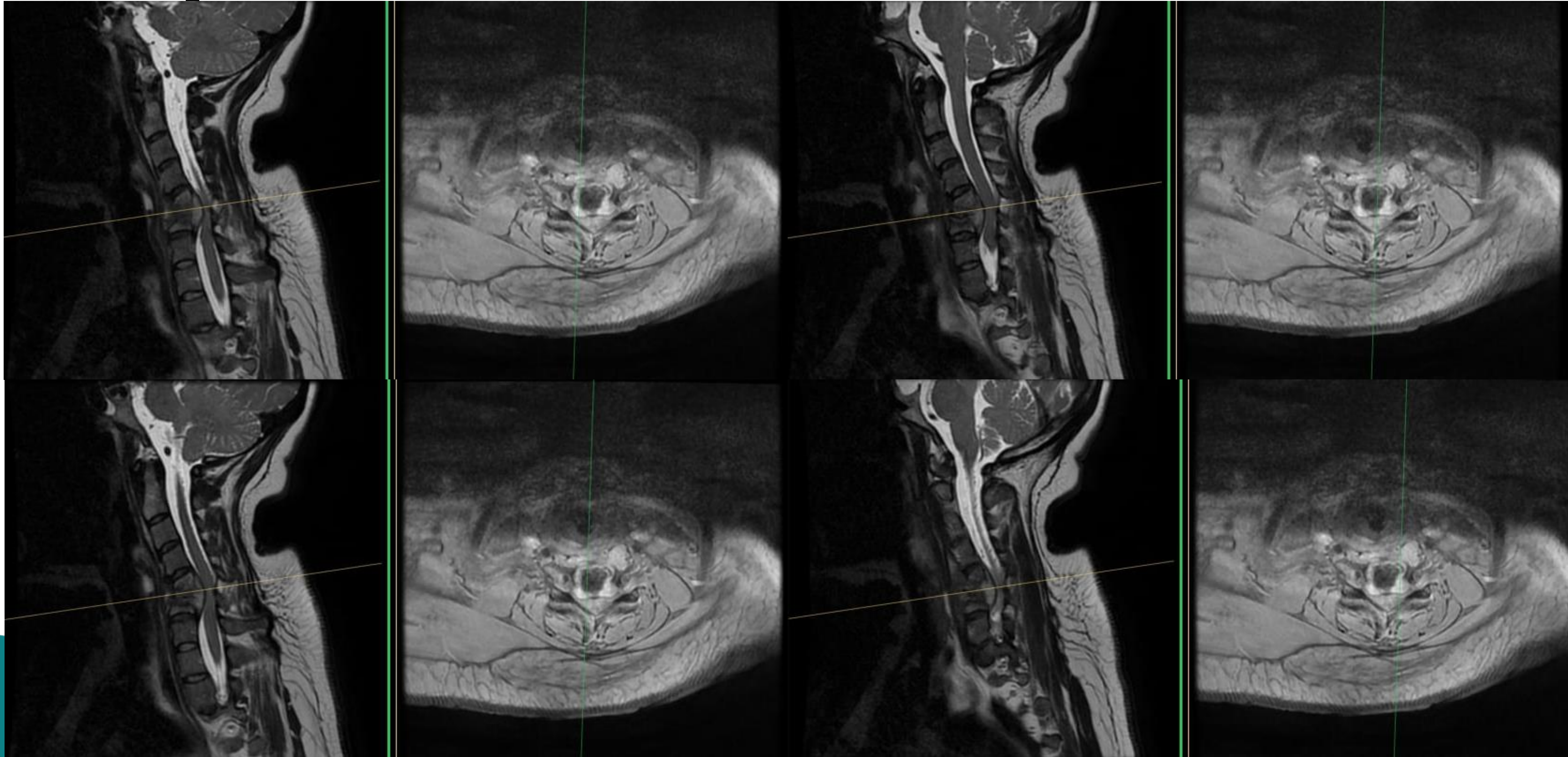
Case Report



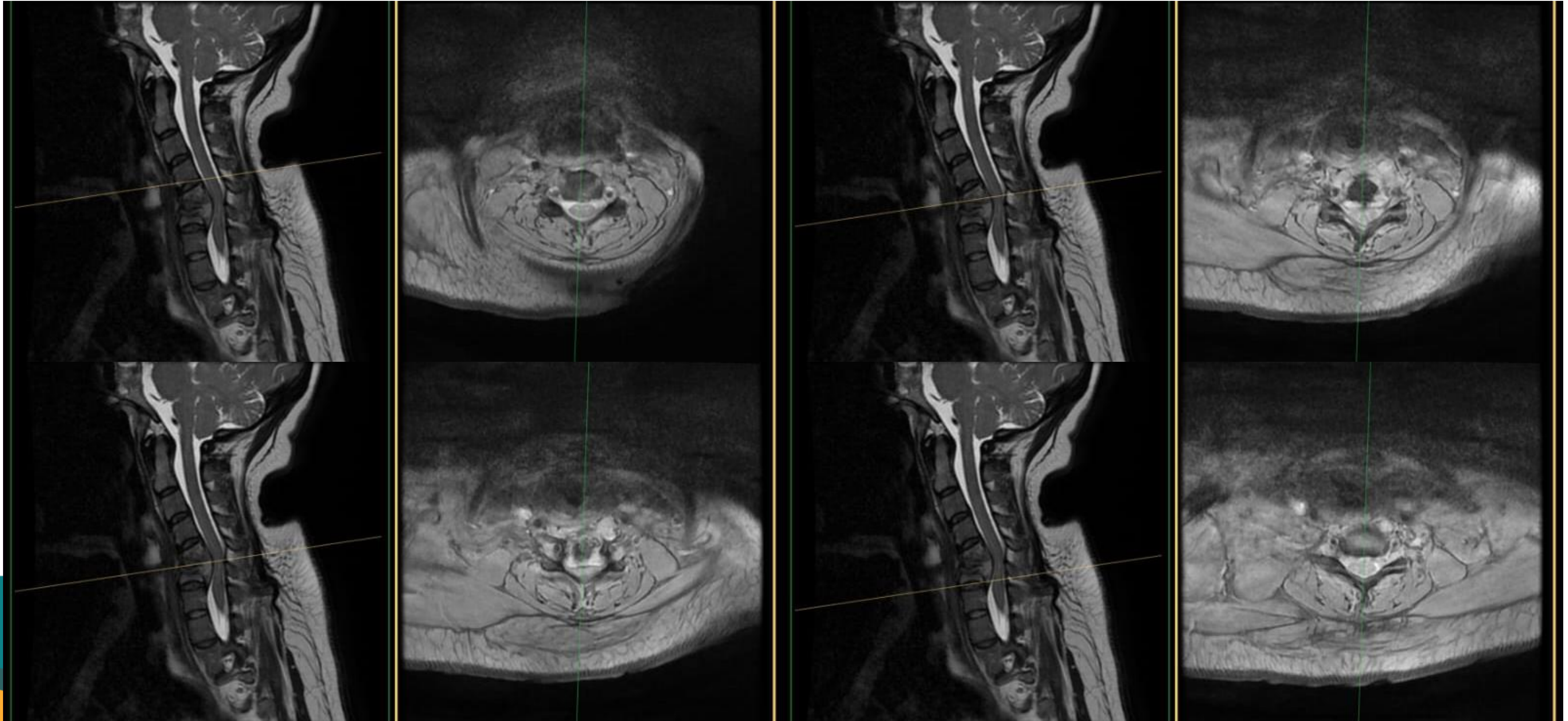
Case Report



Case Report



Case Report



Case Report

Treatment:

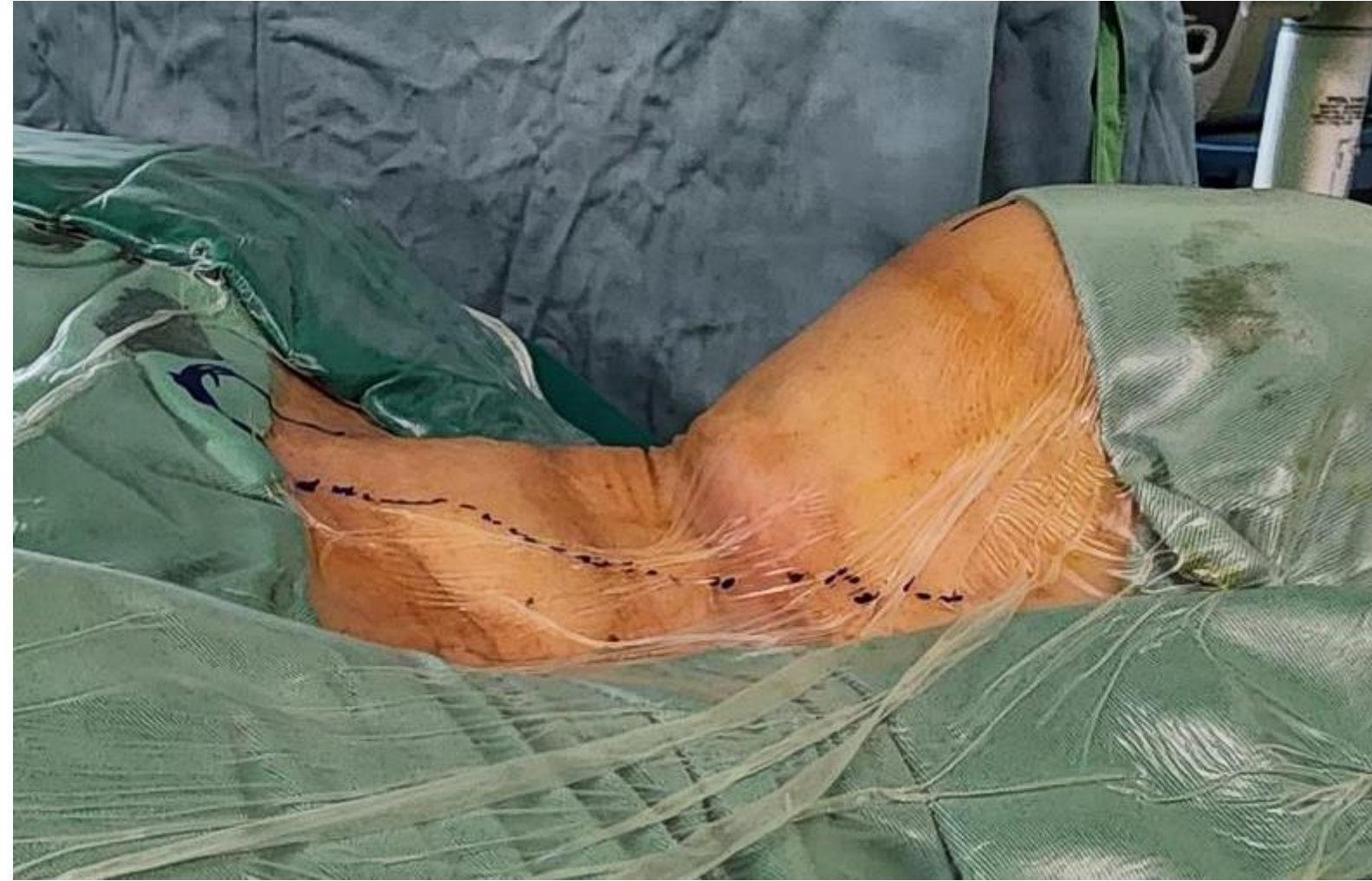
Debridement, biopsy, with Anterior Cervical Corpectomy.

A left-sided Smith-Robinson approach was performed.

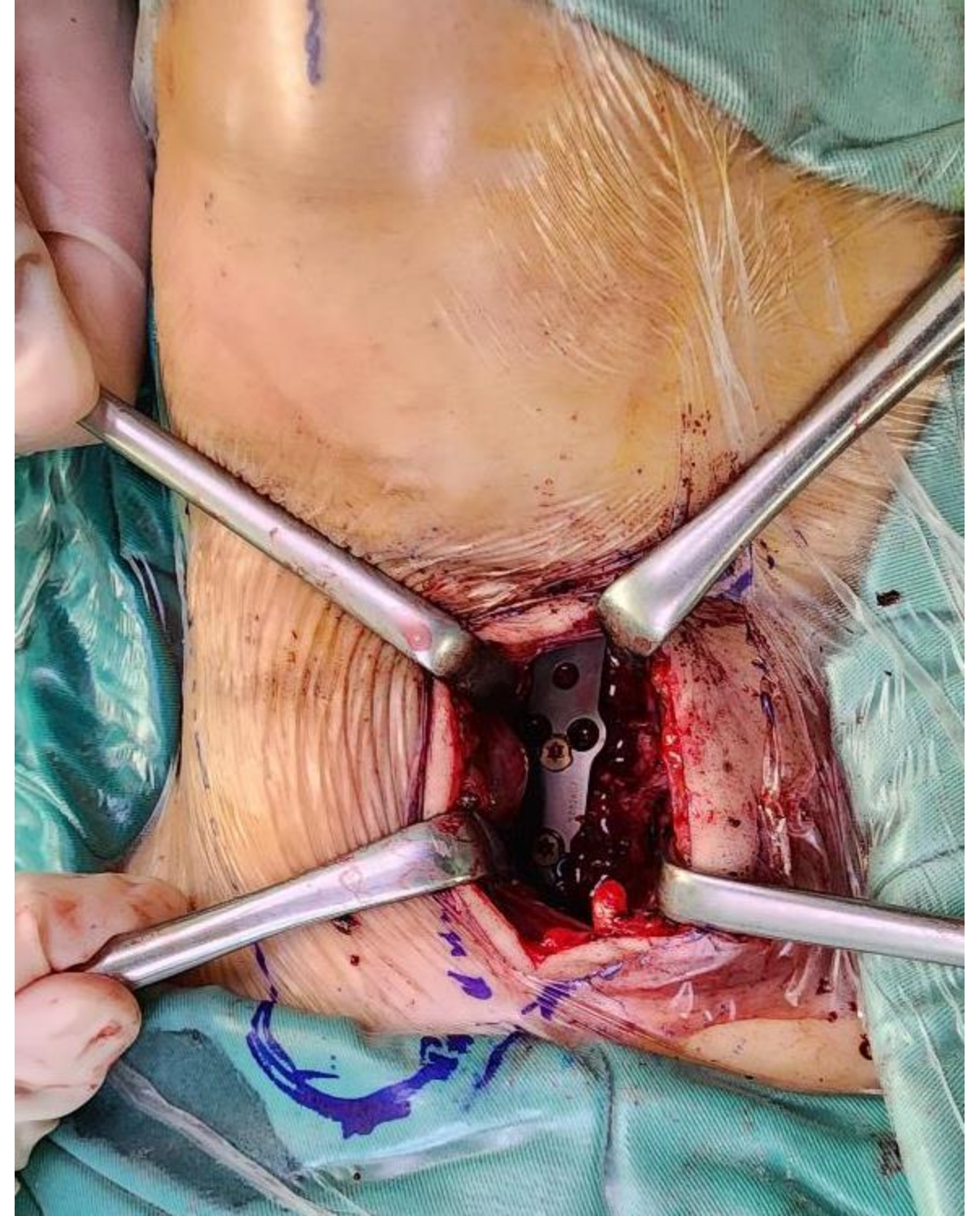
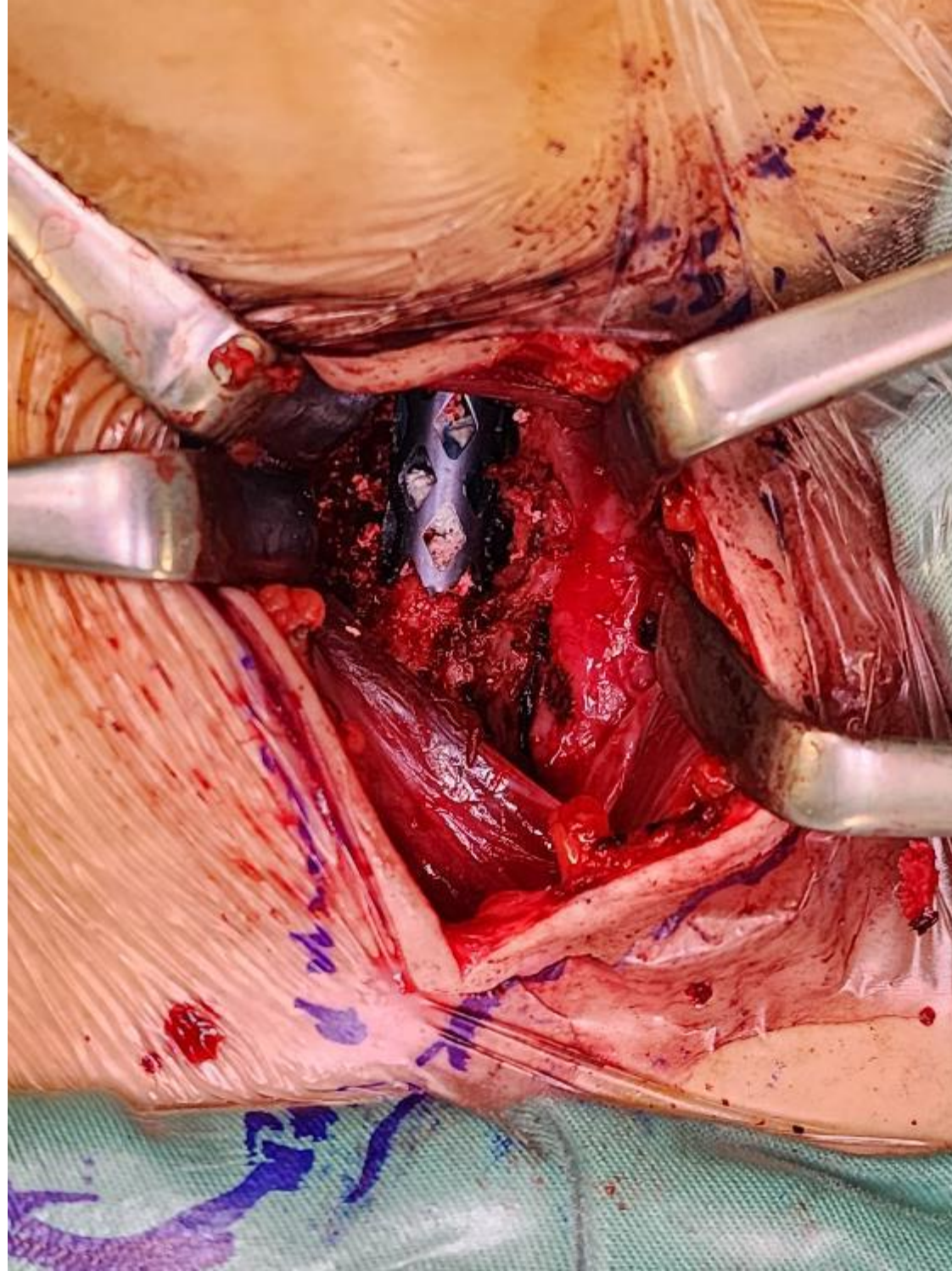
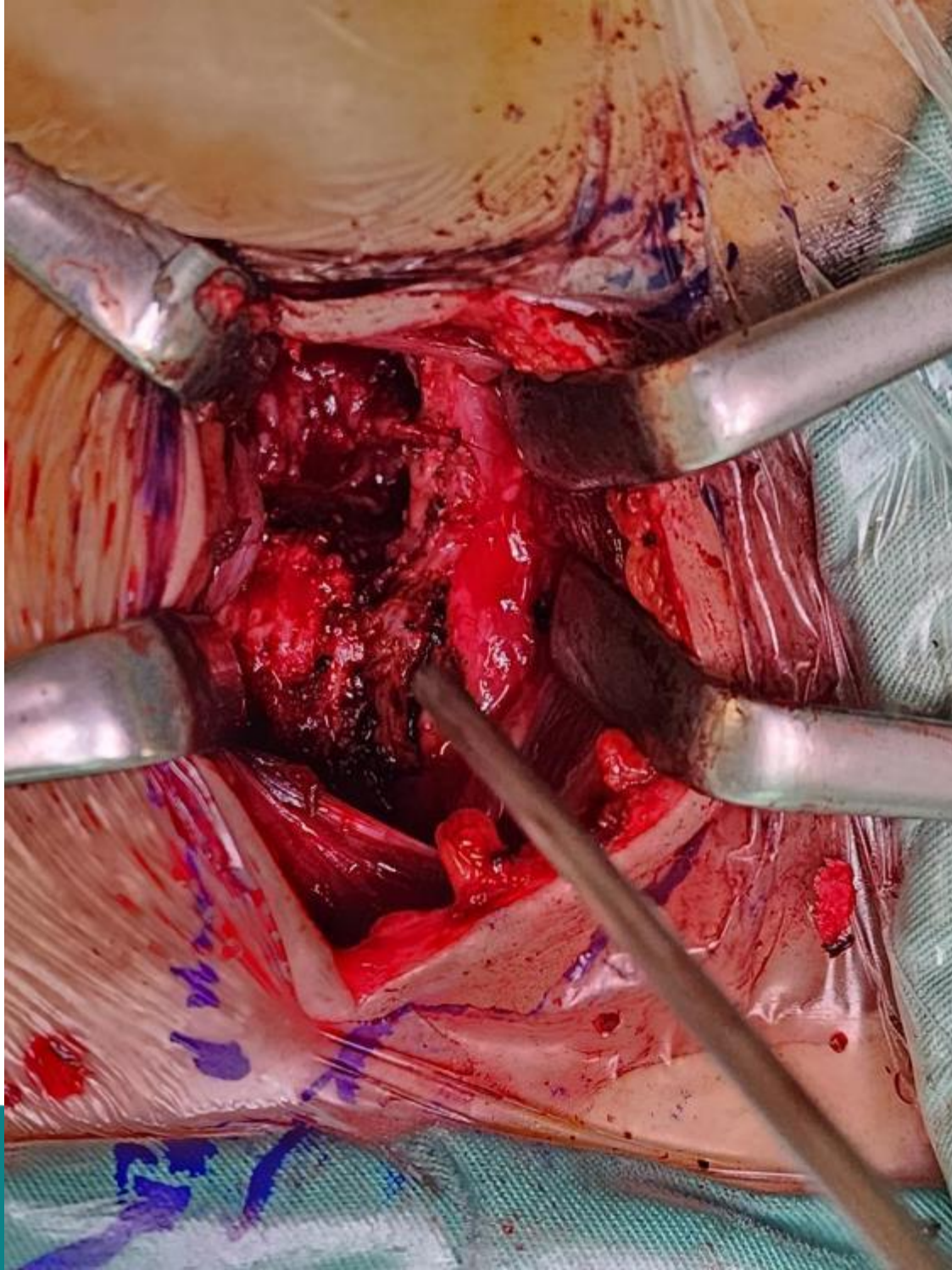


Post surgically, **the patient reported improvements in movement, decreased pain, and no complications.** The patient had no history of tuberculosis.

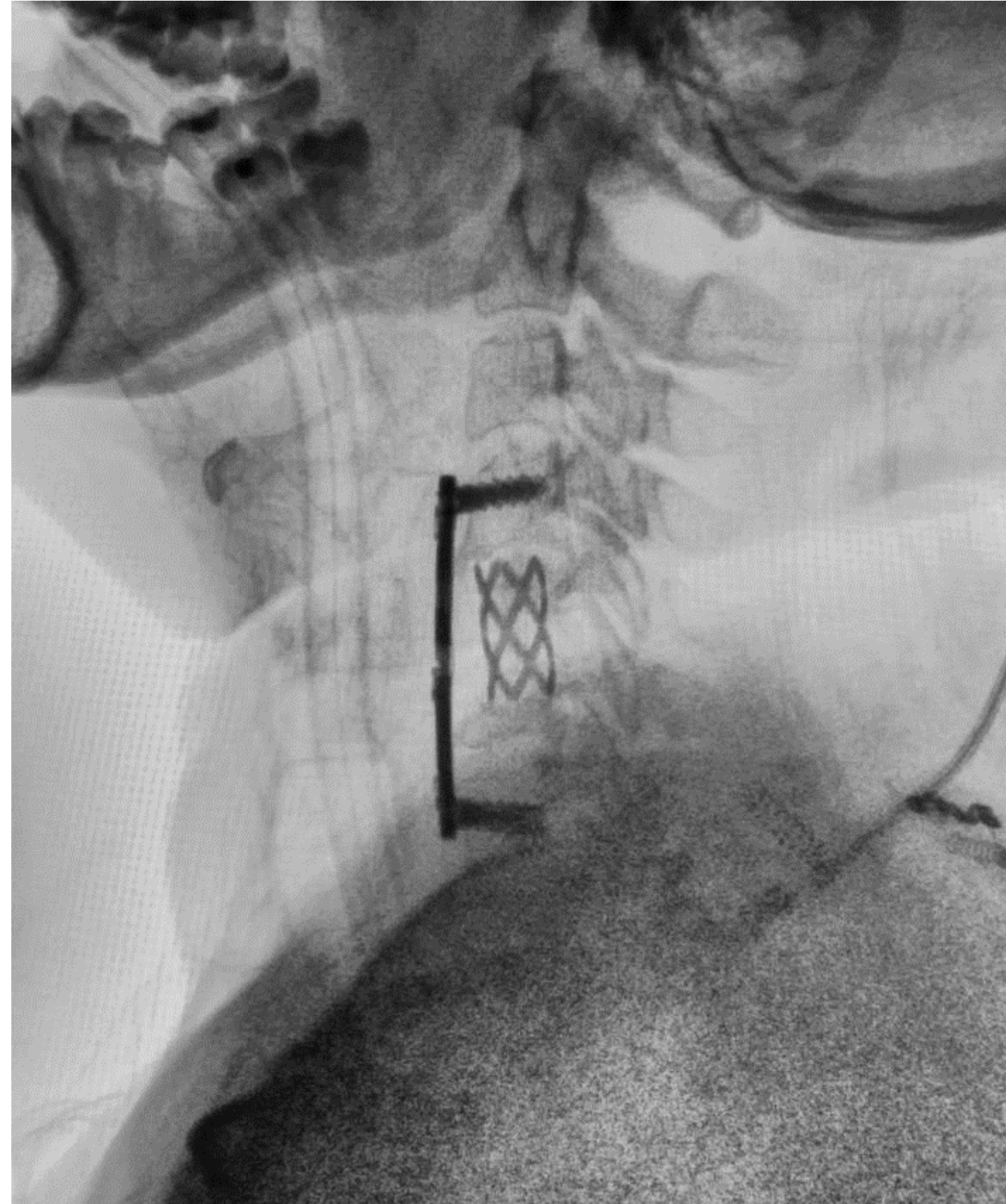
Case Report



Case Report



Case Report



Outcome

Post Op day-1

Motoric:

Shoulder elevation 5/5
Shoulder abduction 5/5
Elbow flexion 5/5
Wrist extension 5/5
Elbow extension 5/5
Wrist flexion 5/5
Finger flexion 5/5
Finger abduction 5/5

Hip flexion 3/4
Knee flexion 3/4
Knee extension 4/4
Ankle dorsiflexion 4/4
Ankle plantarflexion 4/5
Great toe extension 4/5
Great toe flexion 4/5

Sensoric:

Normoesthesia

Physiological reflex:

Biceps: +/+
Triceps: +/+
Patella : +/+
ankle : +/+
Brachioradialis +/+

Pathological reflex:

Babinski -/-
Chaddock -/-
Oppenheim -/-
Clonus -/-

Vegetative:

BAK: on catheter
BAB: normal

Case Report



Discussion

- The left-side approach was chosen, as performed by Dr. Robinson at Johns Hopkins, because **it can minimize the incidence of Recurrent Laryngeal Nerve Palsy (RLNP)** compared to the right approach.

Mak et al.

Stated that the principle of performing the anterior procedure is to recognize the importance of cleaning the paravertebral abscess to achieve perfect healing in spinal tuberculosis.

Robert et al.

This case followed the same procedure performed by Robert et al, with modification, which fusions were performed by creating a trough of bone and filling it in with syntetic chips bone graft.

Hakim et al.

As demonstrated by a case series conducted by Hakim et al, favorable outcomes have also been observed in patients with cervical spinal tuberculosis.

The key distinction between this study and previous research is that the left-sided Smith-Robinson approach resulted in more pronounced improvements, particularly in terms of movement, pain reduction, and a lower incidence of complications.

Conclusion

Operative procedures such as Anterior Cervical Corpectomy and Fusion (ACCF) on cervical spinal tuberculosis cases should be performed when the clinical and radiological findings were clear. Postoperative outcomes using left-sided Smith-Robinson approach were notably positive with no complications.

This case highlights the importance of timely surgical intervention and meticulous postoperative care in achieving favorable clinical outcomes for patients with complex cervical spine pathologies.

Thank you >

